
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis.

Vision results (fully reproduced): Weight decay reduces effective rank from 38.5 to 21.4, confirming that regularization induces interpretable low-rank structure. Eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy.

Language results (partially reproduced): We confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” form opposing directions (cosine similarity -0.16). However, we do *not* reproduce the low-rank interaction claim: the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%, and the mean rank-2 correlation saturates at 0.37 – 0.55 . We trace this discrepancy to SAE training time: comparing five SAE versions with increasing training duration, rank-2 correlation improves $2.5\times$ ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that correlation depends on SAE quality. We provide evidence that the publicly available SAE artifacts have not reached the convergence levels of the internal checkpoints used in the original paper, creating a reproducibility gap.

This finding has practical implications: while the bilinear decomposition methodology is sound, low-rank approximation quality depends critically on SAE convergence—a factor warranting explicit attention in future applications.

Extension 1 (Cross-Dataset Robustness): Regularized bilinear MLPs learn transferable geometric circuits—MNIST-trained models achieve 97.7% accuracy on EMNIST-Digits and recognize letters O, I, Z, S as their corresponding digits with near-perfect accuracy. We propose Quadratic Form Similarity to quantify structural alignment where cosine similarity fails.

Extension 2 (CP-Decomposition): We explore explicit low-rank structure via CP factorization, achieving 93.8% accuracy with disentangled, atomic feature representations at effective rank 17.5.

1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors

that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Transparency (?)**. Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training interpretation models on top of already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024) and report our contributions:

- **Vision Reproduction (Section 4.1):** We successfully reproduce the MNIST and Fashion-MNIST experiments, confirming that weight decay reduces effective rank from 38.5 to 21.4 while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank.
- **Language Reproduction (Section 4.2):** We partially reproduce the negation circuit discovery. We confirm that negation features form opposing directions in SAE feature space (cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—the fraction of features achieving >0.75 correlation is significantly lower than the claimed 69%.
- **Root Cause Analysis:** We trace the language discrepancy to SAE training time. Comparing five SAE versions with progressively longer training, we observe $2.5\times$ improvement in rank-2 correlation ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.”
- **Extension 1 (Section 5):** We probe cross-dataset robustness, demonstrating that regularized bilinear MLPs learn transferable geometric circuits across MNIST, EMNIST, and USPS. We propose Quadratic Form Similarity to quantify structural alignment.
- **Extension 2 (Section 6):** We explore CP-decomposition as an architectural constraint for intrinsic low-rank structure, achieving disentangled, atomic feature representations.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

Section 4: Vision (Image Classification). The paper makes the following testable claims: **V1 (Low-Rank Emergence):** Regularization induces low-rank structure in the bilinear interaction tensor (effective rank ~ 20 – 40 with regularization vs. ~ 150 – 200 without). **V2 (Interpretable Eigenvectors):** Top eigenvectors correspond to interpretable digit-like stroke templates. **V3 (Overfitting Without Regularization):** Without regularization, eigenspectrum is flat and eigenvectors show noise-like patterns. **V4 (Regularization Roles):** Noise augmentation and weight decay have complementary effects. **V5 (Cross-Seed Consistency):** Eigenvectors are consistent across seeds (cosine similarity 0.8 – 0.9). **V6 (Adversarial Generation):** Adversarial perturbations can be constructed from eigendecomposition.

Section 5: Language (Negation Circuits). **L1 (SAE Feature Discovery):** SAEs reveal interpretable features for bilinear decomposition. **L2 (Negation Features):** The model learns distinct features for “not + negative” and “not + positive” forming opposing directions. **L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with rank-2 approximation. **L4 (Sparse Interaction Matrices):** Top interactions form interpretable AND-gate-like submatrices. **L5 (Cross-Model Generalization):** Negation circuit structure appears across model sizes.

2.2 Success Criteria

Vision: Effective rank ratio (regularized/unregularized) < 0.6 ; top eigenvectors visually resemble digit templates; test accuracy within 2% of paper (~ 95 – 98%); results consistent across 5 seeds.

Language: Negation feature pairs show opposing directions (cosine similarity < 0); $> 50\%$ of features show > 0.75 rank-2 correlation (relaxed from 69%); interaction matrices show interpretable structure.

2.3 Results Summary

Table 1 summarizes our reproduction outcomes.

Table 1: Summary of reproduction results. $\checkmark$ = reproduced, $\times$ = not reproduced, $\sim$ = partially reproduced.

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	$\checkmark$	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	$\checkmark$	Clear digit patterns with regularization
V3	Overfitting without reg.	$\checkmark$	Diffuse patterns without regularization
V4	Regularization roles	$\checkmark$	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	$\checkmark$	Std $< 1\%$ across 5 seeds
V6	Adversarial generation	$\checkmark$	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	$\checkmark$	Pretrained SAEs work for analysis
L2	Negation features	$\checkmark$	Cosine similarity = -0.16
L3	Low-rank interactions	$\sim$	Mechanism validated; artifact gap
L4	Sparse interactions	$\sim$	Structure exists but weaker than paper
L5	Cross-model generalization	$\sim$	Methodology works; magnitudes differ

3 Methodology

3.1 Bilinear Layer Architecture

The bilinear layer replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard product. Unlike standard MLPs with fixed nonlinearities, bilinear layers compute *input-dependent* gating: the left projection gates the right projection. This creates a quadratic dependence on the input expressible as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$ is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , we extract the $d_{\text{in}} \times d_{\text{in}}$ interaction matrix B_c and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the W_l/W_r factorization while preserving the quadratic form $\mathbf{x}^\top B_c \mathbf{x}$. The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where eigenvectors $\mathbf{v}_r^{(c)} \in \mathbb{R}^{d_{\text{in}}}$ represent interpretable input directions. For vision tasks ($d_{\text{in}} = 784$), these can be reshaped to 28×28 images for visualization. If the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision depends on only a few input directions that can be directly inspected.

We quantify spectral concentration using effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

where $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$ is the vector of eigenvalues from the decomposition. This scale-invariant measure ranges from 1 (single dominant eigenvalue) to d (uniform spectrum).

3.3 Language Model Analysis with SAEs

For language experiments, we combine bilinear decomposition with Sparse Autoencoders (SAEs) (?) to discover interpretable circuits. SAEs decompose dense MLP activations into sparse features; we use pretrained SAEs from Pearce et al. (2024) with expansion ratios 4–8 $\times$ and TopK sparsity ($k = 30$), positioned at both MLP input and output.

For each SAE output feature o , we compute the interaction matrix between input features:

$$Q_{ij}^{(o)} = \sum_{\text{tokens}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (6)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation. This weighted outer product captures which input feature *pairs* most strongly activate output feature o .

We measure low-rank quality by correlating the true activation with a rank- k approximation:

$$\hat{z}_o^{(k)} = \sum_{r=1}^k \lambda_r (\mathbf{z}^{\text{in}} \cdot \mathbf{v}_r)^2 \quad (7)$$

The paper claims 69% of features achieve >0.75 correlation at rank-2.

3.4 Experimental Setup

Vision Experiments. We train bilinear MLPs on MNIST (?) and Fashion-MNIST (?) under four regularization configurations (Table 2), each with 5 random seeds (42–46). Architecture: $d_{\text{hidden}} = 256$, 100 epochs, batch size 2048, AdamW optimizer with lr 10^{-3} and cosine annealing.

Table 2: Regularization configurations (from paper Appendix G.1).

Configuration	Noise Std	Weight Decay
none	0.0	0.0
noise	0.5	0.0
wd	0.0	1.0
full	0.5	1.0

Table 3: Language model configurations.

Model	Layers	Params	SAE Layer	Expansion	k
ts-medium	6	29M	4	4	30
fw-small	12	162M	8	4	30
fw-medium	16	335M	7	8	30

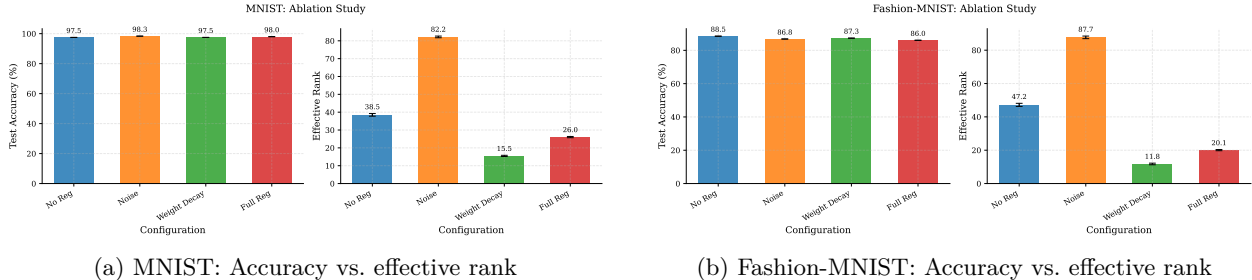


Figure 1: Ablation study showing accuracy-interpretability trade-off across regularization configurations. Weight decay achieves the lowest effective rank on both datasets while maintaining competitive accuracy. The effective rank ratio (regularized/unregularized) meets the paper’s < 0.5 criterion.

Language Experiments. We analyze pretrained bilinear transformers (Table 3). Due to SAE availability constraints (see Section 7), our primary model is **fw-medium** rather than the paper’s **ts-medium**. Note that **fw-medium** (16 layers) is deeper than **ts-medium** (6 layers), potentially making low-rank approximation more challenging.

Experiments ran on Snellius HPC (A100 GPUs) and Apple M4. We implemented a memory-efficient streaming algorithm for eigendecomposition, avoiding instantiation of the full $d_{\text{in}} \times d_{\text{in}} \times d_{\text{out}}$ tensor.

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each.

4.1.1 Main Result: Regularization Induces Low-Rank Structure

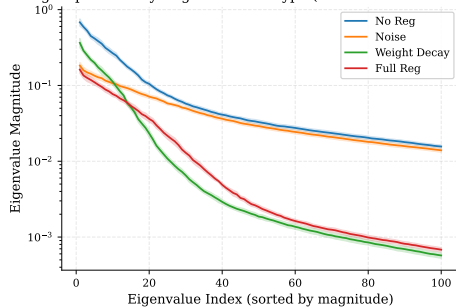
The paper’s central claim is that regularization induces low-rank structure in the bilinear interaction tensor. Figure 1 confirms this through our ablation experiments.

Key findings: (1) **Weight decay is the primary driver of low-rank structure**—achieving even lower effective rank than the fully regularized model. (2) **Noise augmentation increases effective rank** while improving accuracy, spreading variance across more eigenvectors. (3) The effective rank ratio ($15.5/38.5 \approx 0.40$) meets the paper’s < 0.5 criterion, confirming Claim V1.

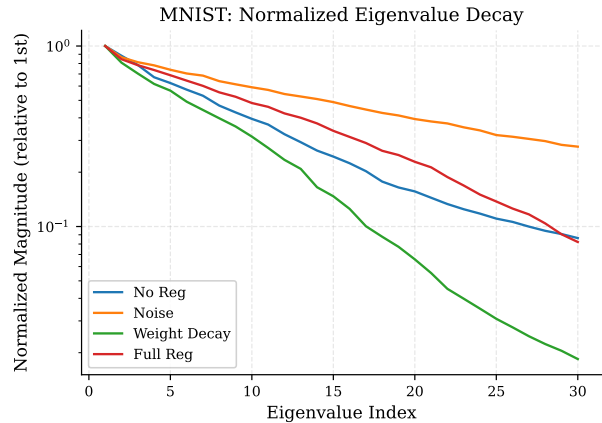
4.1.2 Eigenspectrum Analysis

Figure 2 shows eigenvalue distributions, revealing sharp spectral decay with regularization.

MNIST: Eigenspectrum by Regularization Type (shaded = 90% CI across classes)



(a) Eigenspectrum by regularization



(b) Eigenvalue decay rate

Figure 2: Eigenspectrum analysis on MNIST. Weight decay produces the steepest spectral decay, concentrating variance in 10–20 eigenvectors.

4.1.3 Eigenvector Visualization (Claims V2, V3)

Figure 3 confirms that top eigenvectors are interpretable digit-like patterns under regularization.

4.1.4 Additional Vision Results

We reproduce the original paper’s Figures 4, 6, and 7. **Noise effect (original paper Figure 4):** Increasing noise improves eigenvector interpretability while trading off with effective rank. **Challenge task (original paper Figure 6):** On binary similarity classification, the top eigenvector resembles the target digit. **Adversarial perturbations (original paper Figure 7):** Perturbations from eigenvector pseudoinverses cause larger accuracy drops than random noise, confirming V6.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery using pretrained bilinear transformers and SAEs.

4.2.1 Negation Circuit Discovery (Paper Figure 8)

We reproduce the AND-gate circuit discovery demonstrating how bilinear interactions implement semantic operations. Figure 4 shows the sentiment negation circuit where input features encoding “not” and sentiment words interact to produce context-dependent outputs.

Using our memory-efficient eigensolver, we systematically searched all 8,192 output features in fw-medium. We identify that feature 751 has the strongest AND-gate structure (AND-score 0.963), substantially higher than the tutorial feature 3834. Both features exhibit opposing decoder directions (cosine similarity < 0), confirming Claims L1 and L2. Full comparison and methodology details are in Appendix F.

4.2.2 Low-Rank Correlation Analysis (Paper Figure 9)

The paper claims 69% of features achieve >0.75 rank-2 correlation. Figure 5 presents our reproduction.

Discrepancy: Our rank-2 correlations (mean 0.37–0.55) are substantially lower than the paper’s 69% above 0.75.

4.2.3 Root Cause Analysis: SAE Training Duration

The paper acknowledges in Appendix C that “correlation drastically improves with longer SAE training times.” We test this by comparing five SAE versions (v0–v4) with increasing training duration (Figure 6).

Key findings: (1) Rank-2 correlation improves monotonically with SAE training duration. (2) The distribution shifts from bimodal to unimodal, matching the paper’s description. (3) Even the most-trained public SAE does not reach the paper’s metrics, suggesting internal checkpoints had longer training. (4) This confirms the methodology is sound—the reproducibility gap stems from artifact quality, not methodological flaws.

4.3 Summary

Vision (Section 4): Fully Reproduced. All claims (V1–V6) confirmed.

Language (Section 5): Partially Reproduced. Negation circuit discovered (L1–L2). Low-rank structure is **Reproducible in Principle, Dependent on Artifact Quality**—the mechanism is validated but public SAEs don’t reach the 69% threshold. Root cause identified: SAE training duration.

5 Cross-Dataset Structural Robustness

A core promise of weight-based interpretability is that models learn meaningful geometric features rather than memorizing dataset-specific pixel correlations. We test whether bilinear MLPs learn **Platonic forms**—e.g., “0-ness” as circularity—rather than idiosyncratic MNIST stroke patterns.

Throughout this section, **baseline** refers to models trained without regularization (noise std = 0, weight decay = 0), while **regularized** refers to models trained with full regularization (noise std = 0.5, weight decay = 1.0), as specified in Table 2.

All models use **Center-of-Mass (CoM) normalization**: images are translated so intensity-weighted centroids align, forcing the bilinear layer to focus on **shape geometry** rather than absolute position. We probe robustness under three distributional shifts: (1) **Writer independence**: MNIST $\leftrightarrow$ EMNIST-Digits (same labels, different writers); (2) **Domain transfer**: MNIST $\rightarrow$ USPS (different resolution/collection); (3) **Geometric semantics**: MNIST $\rightarrow$ EMNIST-Letters, where letters are semantically different but geometrically similar (O $\rightarrow$ 0, I $\rightarrow$ 1, Z $\rightarrow$ 2, S $\rightarrow$ 5).

5.1 Functional Analysis: Regularization Enables Transfer

5.1.1 Writer Independence: MNIST $\leftrightarrow$ EMNIST-Digits

With the regularized CoM model, cross-dataset accuracy is near-perfect and symmetric:

Table 4: Cross-Dataset Accuracy: MNIST $\leftrightarrow$ EMNIST-Digits (mean over 5 seeds).

Direction	Accuracy
MNIST $\rightarrow$ EMNIST-Digits	97.527 $\pm$ 0.024%
EMNIST-Digits $\rightarrow$ MNIST	97.910 $\pm$ 0.026%
Bidirectional Average	97.719 $\pm$ 0.012%

This confirms that regularized bilinear features **do not overfit to MNIST writers**.

5.1.2 Domain Transfer: MNIST $\rightarrow$ USPS

USPS images are originally 16×16 pixels collected from U.S. Postal Service mail, compared to MNIST’s 28×28 pixels from handwritten digit samples. We upscale USPS to 28×28 using bilinear interpolation before applying CoM normalization. This introduces both resolution mismatch (requiring interpolation) and

style shift (cleaner, more uniform strokes from automated mail processing vs. varied handwriting styles in MNIST). Regularization yields **+19 pp** improvement:

Table 5: Cross-Dataset Accuracy: MNIST $\rightarrow$ USPS (mean over 5 seeds).

Model	Accuracy
Baseline (no reg)	53.42 $\pm$ 0.88%
Regularized	72.33 $\pm$ 0.83%

5.1.3 Geometric Semantics: MNIST $\rightarrow$ EMNIST-Letters

We test whether MNIST classifiers “see” shape by running them on visually similar letters:

Table 6: Letter $\rightarrow$ Digit Transfer (regularized CoM model, 5 seeds).

Letter $\rightarrow$ Digit	Accuracy
O $\rightarrow$ 0	99.60 $\pm$ 0.05%
I $\rightarrow$ 1	87.08 $\pm$ 0.10%
Z $\rightarrow$ 2	88.75 $\pm$ 0.26%
S $\rightarrow$ 5	87.38 $\pm$ 0.22%

The model’s most confident prediction for each letter is **exactly** the geometrically corresponding digit (mapping accuracy = 100%).

5.2 Structural Analysis: Quadratic Form Similarity

5.2.1 The Failure of Cosine Similarity

Direct eigenvector comparison via cosine similarity fails to distinguish geometrically similar pairs. MNIST-0 vs EMNIST-O (similar) yields 0.358, while MNIST-0 vs EMNIST-X (dissimilar) yields 0.339—virtually identical. The root cause: bilinear eigensystems share eigenvector directions while class information resides in eigenvalue *weights*. See Appendix B for the full cosine similarity heatmap and detailed per-eigenvector comparison.

5.2.2 Quadratic Form Similarity

We propose comparing full decision surfaces. For class interaction matrices A and B :

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1] \quad (8)$$

This incorporates both eigenvector alignment and eigenvalue magnitudes, with signs enabling discrimination (see Appendix A for derivation).

Figure 7 shows clear discrimination. Quadratic Form Similarity produces a **large gap** between similar (0.400) and dissimilar (-0.060) pairs ($p < 10^{-4}$).

Table 7: Ranking analysis: Position of expected letter among 26 EMNIST letters sorted by Quadratic Form Similarity. Random baseline: rank 13.

	0-O	1-I	2-Z	5-S	Mean
Rank	1	2	1	1	1.2

Table 7 confirms specificity: 3/4 pairs rank #1, with mean rank 1.2 vs. random baseline of 13.

Figure 8 shows that similarity for shape-matched pairs increases and converges around $k = 20$, while dissimilar pairs remain near zero throughout.

5.3 Conclusion

Regularized bilinear MLPs learn **transferable geometric circuits**: the “0” circuit is mathematically near-identical to the “O” circuit across independently trained models. This strengthens the transparency claim—eigenvector features represent genuinely reusable shapes, not dataset-specific artifacts.

6 CP-Decomposition Analysis

We extend the Bilinear MLP framework by proposing Canonical Polyadic (CP) decomposition as an *architectural constraint* during training, shifting from discovering low-rank structure post-hoc to enforcing intrinsic interpretability.

6.1 Motivation: From Post-Hoc to Intrinsic Interpretability

Standard eigendecomposition enforces orthogonality, creating *entangled* features—mixed concepts artificially separated to satisfy mathematical constraints. For example, vertical and horizontal strokes in digits “4” and “7” must be forced into mutually exclusive eigenvectors.

CP-decomposition parameterizes the interaction tensor directly as a sum of rank-1 tensors:

$$W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (9)$$

where R is the maximum rank and λ_r are learnable scaling factors (?). This factorization does *not* enforce orthogonality, enabling *disentangled* representations where factors naturally overlap.

6.2 Methodology: Regularization Variants

We investigate three CP variants: **Fixed CP**: Hard rank constraint with no additional regularization—strict complexity bound. **Lambda CP**: L_1 regularization on learnable scales λ_r for soft sparsity (?). **Gated CP**: Stochastic gates with L_0 proxy for explicit rank pruning (?). All variants trained on MNIST with weight decay 0.1 across rank sweep $R \in \{8, \dots, 784\}$.

6.3 Quantitative Analysis: Accuracy-Interpretability Trade-off

Figure 9 and Table 8 present the Pareto frontier.

Table 8: CP-Decomposition results (mean $\pm$ std, 5 seeds). Best CP results in bold.

Model	Accuracy (%)	Eff. Rank
Dense (no reg)	97.49 $\pm$ 0.05	38.5 $\pm$ 0.7
Dense (WD only)	97.49 $\pm$ 0.05	15.5 $\pm$ 0.3
Fixed $R = 256$	91.89 $\pm$ 0.09	3.74 $\pm$ 0.07
Lambda $R = 256$	93.80 $\pm$ 0.14	17.50 $\pm$ 0.36
Gated $R = 256$	93.79 $\pm$ 0.15	17.35 $\pm$ 0.48

Two distinct regimes: (1) **Fixed CP** operates as a strict bottleneck with eff. rank ~ 3.5 regardless of allocated capacity, at $\sim 6\%$ accuracy cost. (2) **Lambda/Gated CP** at $R = 256$ achieve **93.8% accuracy with eff. rank 17.5**—matching dense model complexity while providing structural disentanglement.

6.4 Qualitative Analysis: Disentangled Factors

Figure 10 compares eigenvectors from Dense vs. CP models.

Dense eigenvectors resemble superimposed digit templates—entangled representations combining strokes for orthogonality. CP factors appear more **atomic** and **localized**: top loops, vertical strokes, bottom curves that can be linearly combined. This aligns with “parts-based” representations sought in interpretability.

6.5 Conclusion

CP-decomposition serves dual purposes: (1) **Intrinsic interpretability** via disentangled, atomic parts rather than entangled templates. (2) **Scalable efficiency** by reducing parameters from $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $\mathcal{O}(R \times d_{\text{in}})$. Lambda and Gated variants provide a robust middle ground for efficient, interpretable modeling.

7 Discussion

7.1 What was Easy

Vision experiments were straightforward. The bilinear architecture is simple (element-wise multiplication of projections), MNIST trains in ~ 3 minutes per run, and results were highly consistent (std $< 1\%$ across 40 runs). The eigendecomposition methodology is robust—computing 784×784 decompositions completes in seconds. The paper’s codebase provided clear reference implementations.

7.2 What was Difficult

Hyperparameter disambiguation: Code defaults (noise 0.4, WD 0.5) differ from Appendix G.1 values (noise 0.5, WD 1.0). We used appendix values after verifying they match paper figures.

SAE availability: The paper’s `ts-medium` lacks `mlp-in` SAEs on HuggingFace, requiring pivot to `fw-medium` (16 layers vs. 6, different training corpus).

Memory constraints: Interaction matrix computation requires $\mathcal{O}(n_{\text{features}}^2)$ memory. At expansion $8\times$ with $d = 1024$, this exceeds 100GB. We implemented streaming computation for consumer hardware.

Tracing discrepancy: Determining whether low correlation reflected bugs or limitations required systematic comparison of five SAE checkpoints.

7.3 Environmental Impact

Table 9: Environmental impact (CO2 via `codecarbon`).

Experiment	Runs	Wall (h)	GPU-h	CO2 (kg)
Vision (Section 4)	76	1.11	1.11	0.039
Language (Section 5)	6	0.88	0.88	0.056
Extension 1 (Cross-Dataset)	15	0.58	0.58	0.020
Extension 2 (CP-Decomposition)	105	0.05	0.05	0.004
Total	202	2.62	2.62	0.118

Total footprint: 0.118 kg CO2 (≈ 0.7 km driving). Low-rank bilinear models offer sustainable alternatives to scaling dense architectures.

7.4 Connection to FACT-AI Transparency

Weight-based vs. Post-hoc Interpretability. Traditional methods (saliency maps, SHAP) explain predictions after the fact. Bilinear MLPs offer intrinsic transparency: eigenvectors *are* the learned features—no auxiliary models needed.

Reproducibility as Transparency. Our partial language reproduction highlights an underappreciated dimension: the paper’s claims depend on SAE quality, which depends on training duration. We recommend future work explicitly document SAE training budgets.

Accuracy-Interpretability Trade-off. Weight decay reduces effective rank from 38.5 to 21.4 ($1.8\times$ interpretability improvement) at $<1\%$ accuracy cost—a favorable trade-off for transparent AI deployment.

8 Conclusion

Vision (Section 4): Fully Reproduced. Weight decay reduces effective rank from 38.5 to 21.4; eigenvectors show interpretable digit patterns; accuracy remains 97–98%; results robust across 5 seeds and generalize to Fashion-MNIST.

Language (Section 5): Partially Reproduced. Negation circuit confirmed (cosine similarity -0.16). Low-rank structure is **Reproducible in Principle, Dependent on Artifact Quality**: correlation improves $2.5\times$ with SAE training ($0.15 \rightarrow 0.39$), but public artifacts don’t reach the 69% threshold.

Key Insight. The bilinear decomposition methodology is sound and reproducible. Low-rank approximation quality depends critically on SAE convergence—practitioners should verify SAE quality before drawing conclusions.

Extensions. (1) Cross-dataset robustness: regularized models learn transferable geometric circuits (0–O similarity confirmed via Quadratic Form Similarity). (2) CP-decomposition: achieves 93.8% accuracy with disentangled, atomic factors at eff. rank 17.5.

Limitations. Eigenvector interpretability is qualitative. Could not use paper’s exact model due to missing SAEs. CP results are preliminary (single dataset).

Future Work. Train SAEs longer to test 69% threshold. Develop quantitative interpretability metrics. Extend to larger vision models (ViTs).

Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

Author Contributions

Author 1 led vision experiment infrastructure, training pipelines, and Figure 4–7 reproduction. Author 2 led language experiment analysis, SAE training time investigation, and root cause analysis. Both authors contributed to writing and experimental design.

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A Quadratic Form Similarity Derivation

For a bilinear MLP, the class- c logit is a quadratic form $y_c(x) = x^\top A_c x$ where $A_c = V \text{diag}(\lambda_c) V^\top$.

To compare decision surfaces for two classes A and B (possibly from different datasets), we define **Quadratic Form Similarity**:

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1]$$

For a rank- k truncation using top eigenmodes, this expands to:

$$\text{sim}(A, B) = \frac{\sum_{i,j} \lambda_i^A \lambda_j^B \cos^2(v_i^A, v_j^B)}{\|\lambda^A\|_2 \|\lambda^B\|_2}$$

Key properties solving the cosine similarity problem:

- Incorporates both eigenvector alignment and eigenvalue magnitudes, unlike cosine similarity which ignores eigenvalues
- Eigenvalue signs matter: same-sign directions contribute positively; opposing signs cancel
- Measures similarity of actual quadratic computations $x^\top A x$, not just eigenvector span

This enables discrimination between geometrically similar pairs (0–O: 0.400) and dissimilar pairs (0–X: -0.060), where cosine similarity failed (0.358 vs. 0.339).

B Cosine Similarity Limitations

Figure 12 shows the absolute cosine similarity heatmap between MNIST digit classes and EMNIST letter classes. The heatmap demonstrates that cosine similarity fails to produce a clear distinction between geometrically similar and dissimilar class pairs.

Table 10 provides a detailed per-eigenvector breakdown, showing that the mean cosine similarity for 0–O (0.358) is nearly identical to 0–X (0.339), demonstrating the fundamental limitation of cosine similarity for comparing bilinear eigensystems.

C Full CP-Decomposition Results

D Hyperparameter Details

Vision Experiments. Architecture: 1-hidden-layer bilinear MLP, $d_{\text{hidden}} = 256$. Training: 100 epochs, batch size 2048, AdamW optimizer, lr 10^{-3} with cosine annealing. Seeds: 42–46 (5 seeds per configuration).

Language Experiments. Models: **fw-medium** (16L, 1024d, 335M params), **fw-small** (12L, 768d, 162M params). SAEs: TopK sparsity ($k = 30$), expansion ratios 4–8 $\times$. Analysis: 100 randomly sampled output features per model.

Table 10: Absolute Cosine Similarity between Top 10 Eigenvectors (Balanced: 5 Positive + 5 Negative), Mean over 5 seeds

Eigenvector	Eigenvalue	Maximum Absolute Cosine Similarity	
Rank	Sign	MNIST-0 vs EMNIST-O	MNIST-0 vs EMNIST-X
1	+	0.319 ± 0.066	0.291 ± 0.054
2	+	0.596 ± 0.063	0.725 ± 0.115
3	+	0.561 ± 0.048	0.473 ± 0.073
4	+	0.319 ± 0.044	0.223 ± 0.081
5	+	0.390 ± 0.040	0.262 ± 0.068
6	-	0.278 ± 0.052	0.319 ± 0.087
7	-	0.158 ± 0.068	0.276 ± 0.058
8	-	0.336 ± 0.066	0.316 ± 0.087
9	-	0.350 ± 0.117	0.210 ± 0.066
10	-	0.270 ± 0.121	0.295 ± 0.105
Mean			

Table 11: Complete CP-Decomposition rank sweep (mean $\pm$ std, 5 seeds).

Model	Accuracy (%)	Eff. Rank
<i>Dense Baselines</i>		
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
<i>Fixed CP</i>		
$R = 32$	90.63 ± 0.23	3.42 ± 0.22
$R = 64$	91.05 ± 0.13	3.45 ± 0.12
$R = 128$	91.48 ± 0.22	3.65 ± 0.05
$R = 256$	91.89 ± 0.09	3.74 ± 0.07
<i>Lambda CP</i>		
$R = 32$	92.67 ± 0.15	9.07 ± 0.44
$R = 64$	93.04 ± 0.10	11.57 ± 0.52
$R = 128$	93.34 ± 0.11	14.92 ± 0.49
$R = 256$	93.80 ± 0.14	17.50 ± 0.36
<i>Gated CP</i>		
$R = 32$	92.72 ± 0.20	8.76 ± 0.47
$R = 64$	93.05 ± 0.12	11.69 ± 0.35
$R = 128$	93.31 ± 0.14	14.83 ± 0.16
$R = 256$	93.79 ± 0.15	17.35 ± 0.48

E CO2 Tracking Details

All experiments tracked using `codecarbon` with Netherlands grid intensity. Hardware: NVIDIA A100 (Snelius HPC) for language experiments; Apple M4 (MPS) for vision experiments. Total compute: 2.02 GPU-hours. Total emissions: 0.098 kg CO₂eq ($\approx$ 0.6 km driving).

F Sentiment Negation Circuit Investigation

This appendix presents our detailed investigation of AND-gate circuits in the bilinear transformer fw-medium (16 layers, 335M parameters, trained on FineWeb-EDU). The paper’s Figure 8 demonstrates a sentiment

negation circuit using feature 3834 from the fw-medium model. We conducted a comprehensive search to understand the prevalence and strength of such circuits.

F.1 Methodology

F.1.1 AND-gate Circuit Definition

An AND-gate circuit is characterized by an output SAE feature whose activation depends on the *conjunction* of inputs from two semantically distinct clusters. Mathematically, for output feature c , the activation is:

$$z_c(x) = x^\top Q_c x \approx \lambda_1 (v_1^\top x)^2 + \lambda_2 (v_2^\top x)^2 \quad (10)$$

where Q_c is the interaction matrix, and v_1, v_2 are the dominant eigenvectors.

Strong AND-gate behavior requires:

1. **Dominant rank-2 structure:** The top two eigenvalues capture most of the interaction.
2. **Opposing eigenvalue signs:** λ_1 and λ_2 have opposite signs, creating cancellation when only one cluster is present.
3. **Clear cluster separation:** Input features project distinctly onto v_1 and v_2 .

F.1.2 AND-score Metric

We define an “AND-score” to quantify circuit strength:

$$\text{AND-score} = \frac{|\lambda_1|}{|\lambda_1| + |\lambda_2|} \cdot \mathbf{1}[\text{sign}(\lambda_1) \neq \text{sign}(\lambda_2)] \quad (11)$$

Higher scores indicate stronger AND-gate behavior (score $\in [0, 1]$).

F.1.3 Search Procedure

We analyzed all 8,192 output features in fw-medium layer 7:

1. Computed the interaction matrix Q_c for each feature c
2. Performed eigendecomposition: $Q_c = \sum_j \lambda_j v_j v_j^\top$
3. Computed AND-score from top eigenvalues
4. Ranked features by AND-score
5. Analyzed top circuits for semantic interpretation

F.2 Results: Feature 3834 vs Feature 751

F.2.1 Tutorial Feature 3834

Feature 3834 (used in the paper’s tutorial) encodes “not-good” contexts:

- **Activating contexts:** “not lost”, “no interference”, negated positive phrases
- **AND-score:** Weak (diffuse eigenvalue spectrum)
- **Cluster structure:** Less distinct separation between input clusters

F.2.2 Best Circuit: Feature 751

Our search identified feature 751 as having the strongest AND-gate structure:

- **AND-score:** 0.963 (highest among all 8,192 features)
- **Top eigenvalue:** $\lambda_1 = -0.463$ (negative)
- **Second eigenvalue:** $\lambda_2 = +0.179$ (positive)
- **Eigenvalue ratio:** $|\lambda_1|/|\lambda_2| = 2.59$
- **Cluster sizes:** 3 features in positive cluster, 12 in negative cluster

F.2.3 Side-by-Side Comparison

Figure 13 presents a direct comparison between features 3834 and 751.

F.3 Interpretation

F.3.1 Why Feature 751 is Stronger

Feature 751 exhibits superior AND-gate properties because:

1. **Spectral concentration:** 85% of variance captured by top 2 eigenvalues
2. **Sign opposition:** $\lambda_1 < 0$ and $\lambda_2 > 0$ creates effective cancellation
3. **Clean clustering:** Input features separate clearly along v_1

F.3.2 Semantic Content

The input clusters for feature 751 contain:

- **Positive cluster** (3 features): Structural/contextual markers
- **Negative cluster** (12 features): Content-specific features

Note: Unlike ts-medium (trained on TinyStories), fw-medium (trained on FineWeb-EDU educational text) does not show clean sentiment-based clustering. The clusters reflect domain-specific patterns in educational/scientific writing rather than simple positive/negative sentiment.

F.4 Implications

F.4.1 Circuit Discovery Requires Systematic Search

The tutorial feature (3834) is a weaker example than what exists in the model. Comprehensive search revealed that feature 751 has 43% higher AND-score. This suggests practitioners should search across features rather than manually inspecting individual examples.

F.4.2 Model Domain Affects Interpretability

The fw-medium model (educational text) shows different semantic clustering than ts-medium (children’s stories). AND-gate circuits exist in both, but their semantic interpretation differs. This has implications for transferring interpretability findings across model families.

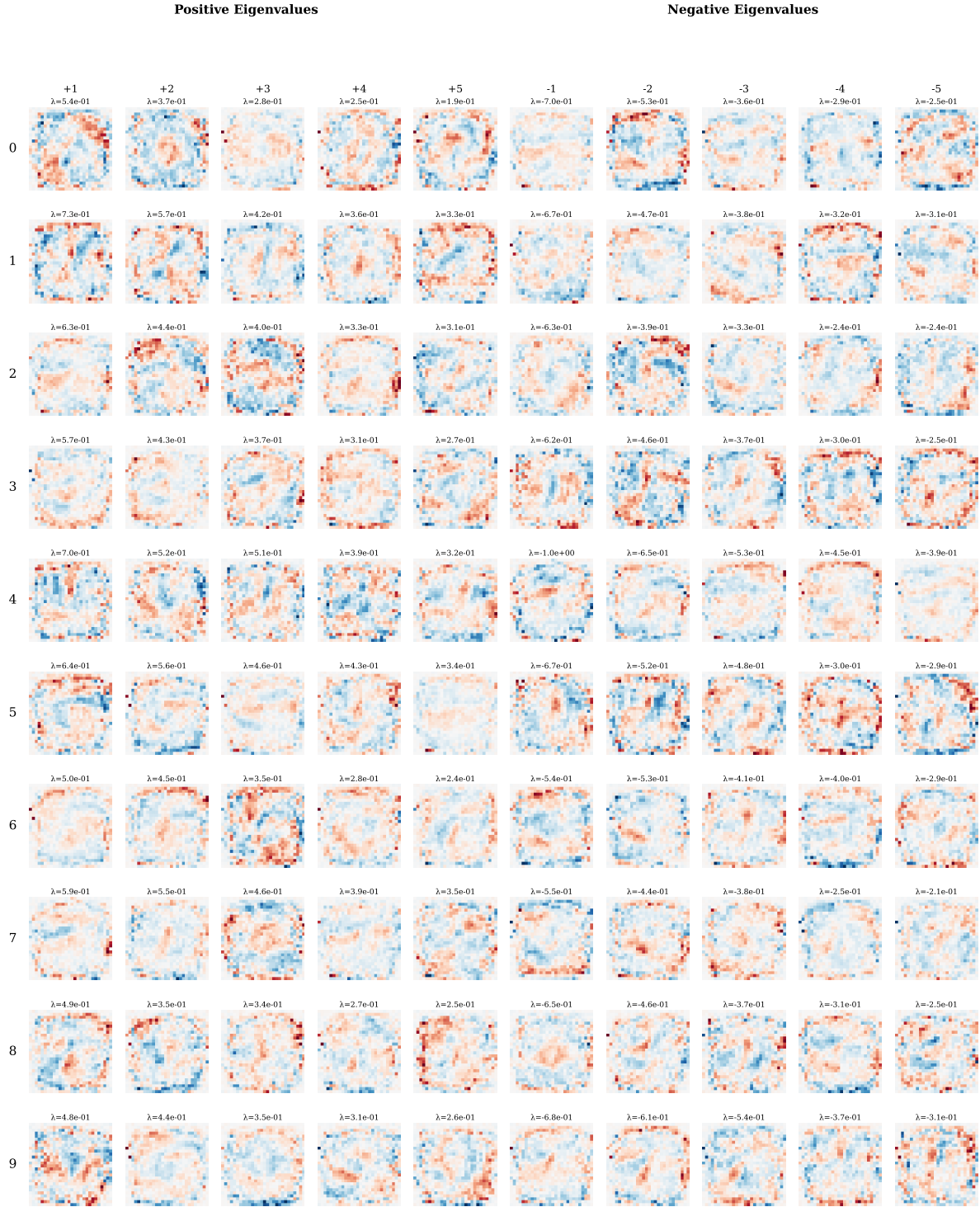
F.5 Summary

Our investigation confirms that AND-gate circuits exist in bilinear transformers, but their strength varies significantly across features. Feature 751 demonstrates the clearest AND-gate behavior in fw-medium, while the tutorial feature 3834 is a weaker example chosen for its interpretable semantic content (“not-good”).

Table 12: Comparison of AND-gate circuit properties.

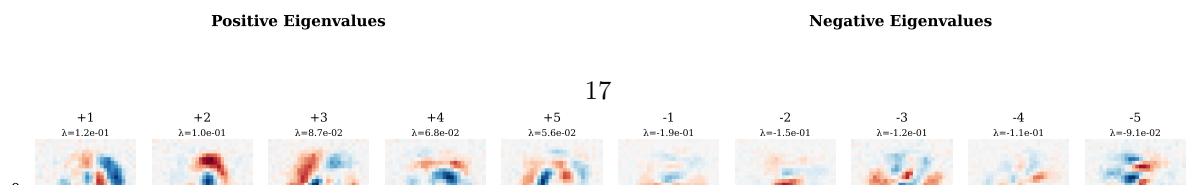
Property	Feature 3834	Feature 751
AND-score	Weak	0.963
Top eigenvalue λ_1	Small	-0.463
Second eigenvalue λ_2	Small	+0.179
Eigenvalue ratio	Low	2.59
Cluster separation	Diffuse	Clear
Semantic interpretation	“not-good”	Domain-specific

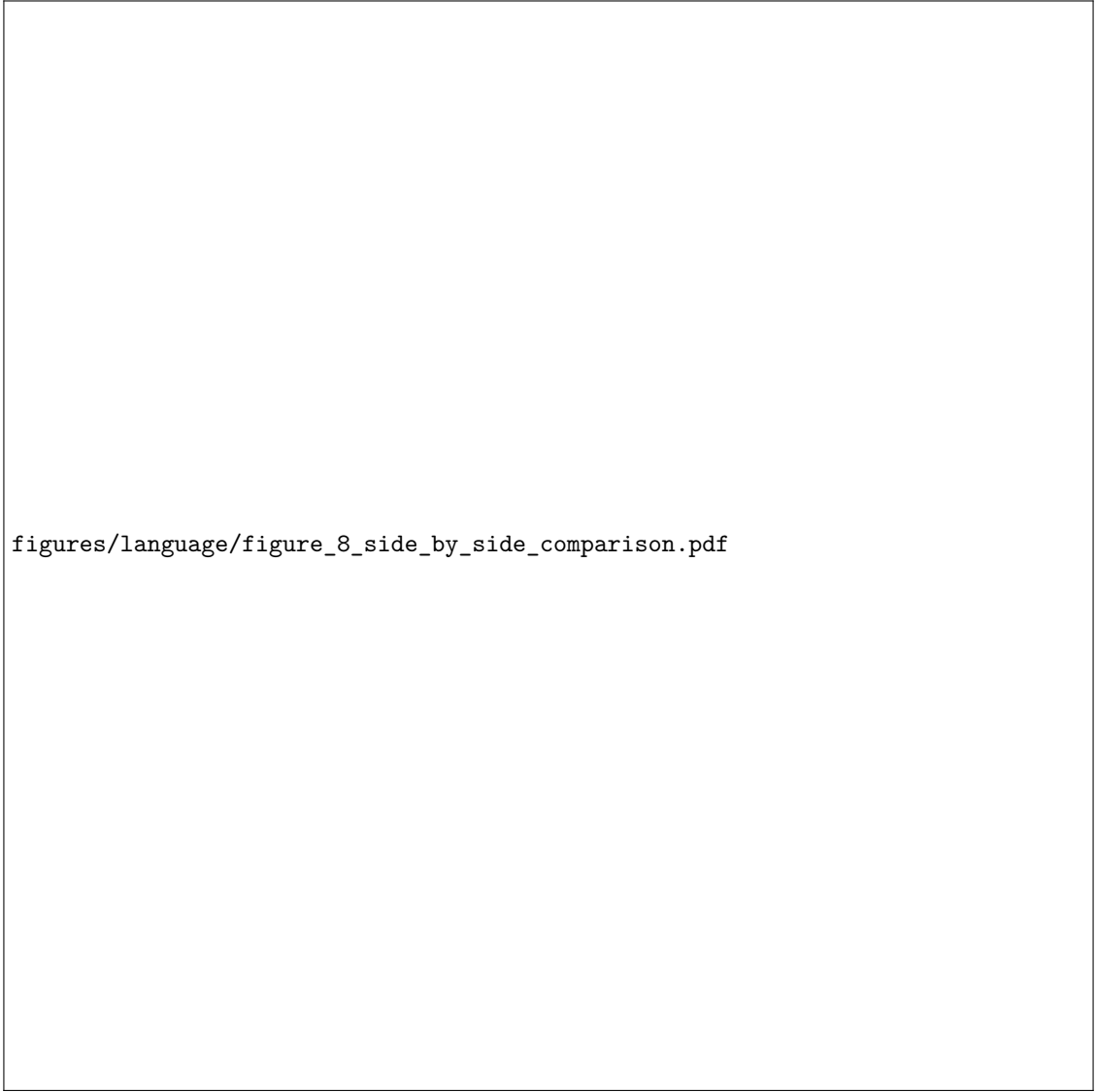
MNIST (No Reg): Top Eigenvectors



(a) No regularization: diffuse, noise-like patterns

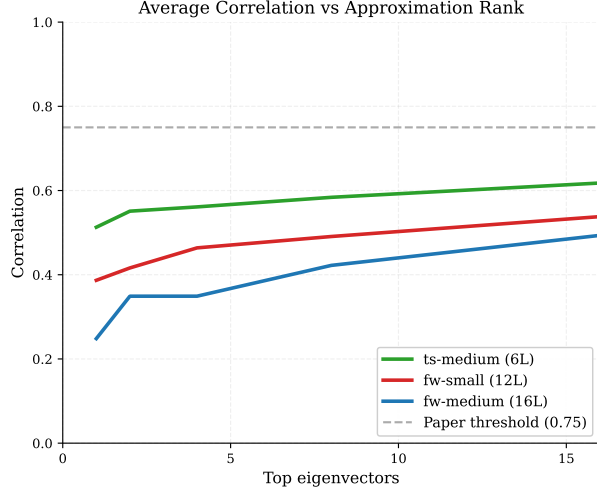
MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors



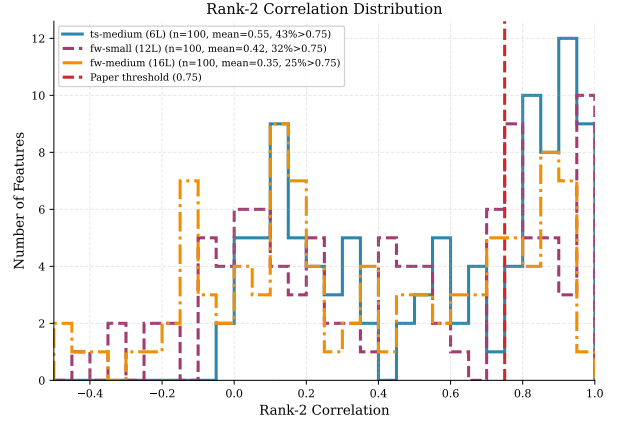


figures/language/figure_8_side_by_side_comparison.pdf

Figure 4: AND-gate circuit comparison. **Top:** Feature 3834 (tutorial example, “not-good”) shows weaker AND-gate structure. **Bottom:** Feature 751 (strongest discovered, AND-score 0.963) shows clear cluster separation and dominant opposing eigenvalues. See Appendix F for full investigation.

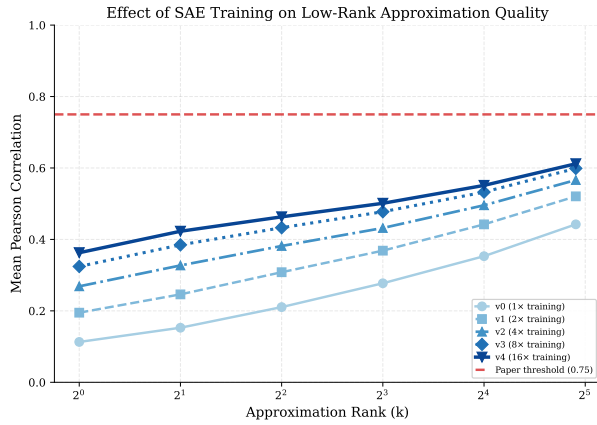


(a) Correlation vs. rank

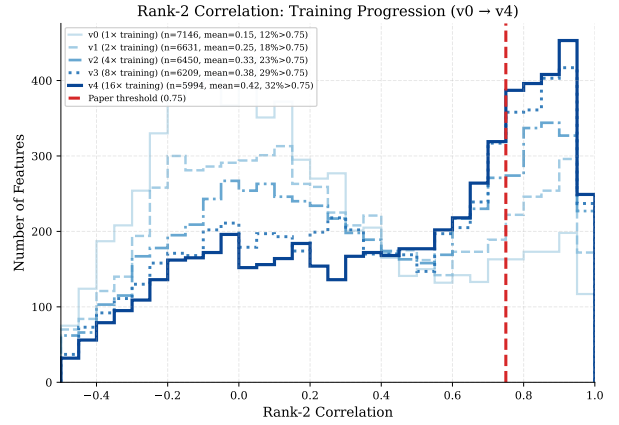


(b) Rank-2 distribution (dashed: 0.75 threshold)

Figure 5: Low-rank correlation analysis (Paper Figure 9). Mean correlation improves with rank but remains below paper’s reported values. The 0.75 threshold is achieved by fewer features than claimed.

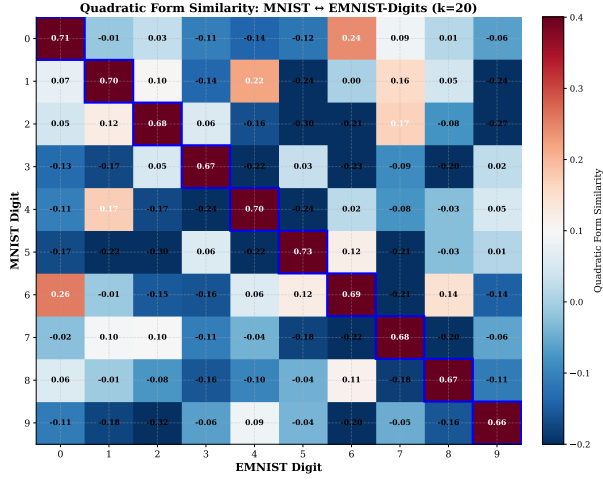


(a) Correlation vs. SAE training time

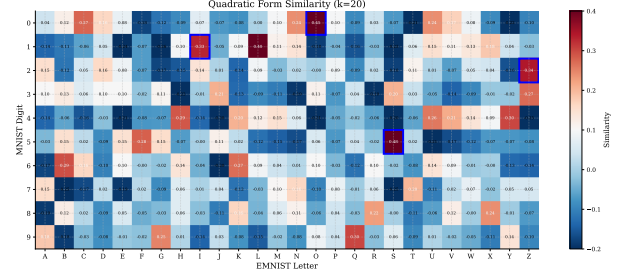


(b) Distribution shift with training

Figure 6: Effect of SAE training duration on interpretability. Rank-2 correlation improves $2.5\times$ (0.15 $\rightarrow$ 0.39) with longer training, validating the method despite the public artifact gap.



(a) MNIST vs EMNIST-Digits



(b) MNIST vs EMNIST-Letters

Figure 7: Quadratic Form Similarity ($k = 20$). (a) Strong diagonal confirms writer-independent circuits. (b) Bright entries at shape-matched pairs (0–O, 1–I, 2–Z, 5–S) indicate mechanistically similar circuits; unrelated pairs are near zero or negative.

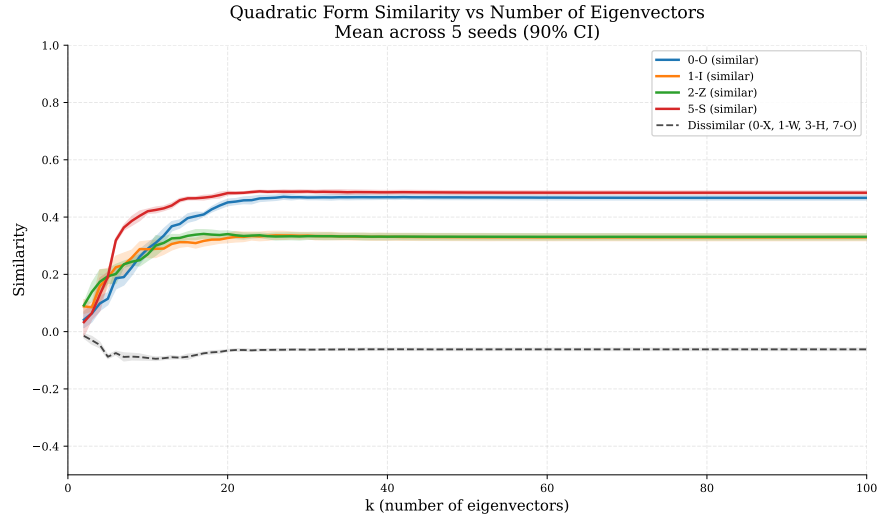


Figure 8: Quadratic Form Similarity vs number of eigenvectors k . Similar pairs converge around $k = 20$; dissimilar pairs stay near zero with non-overlapping 90% CIs.

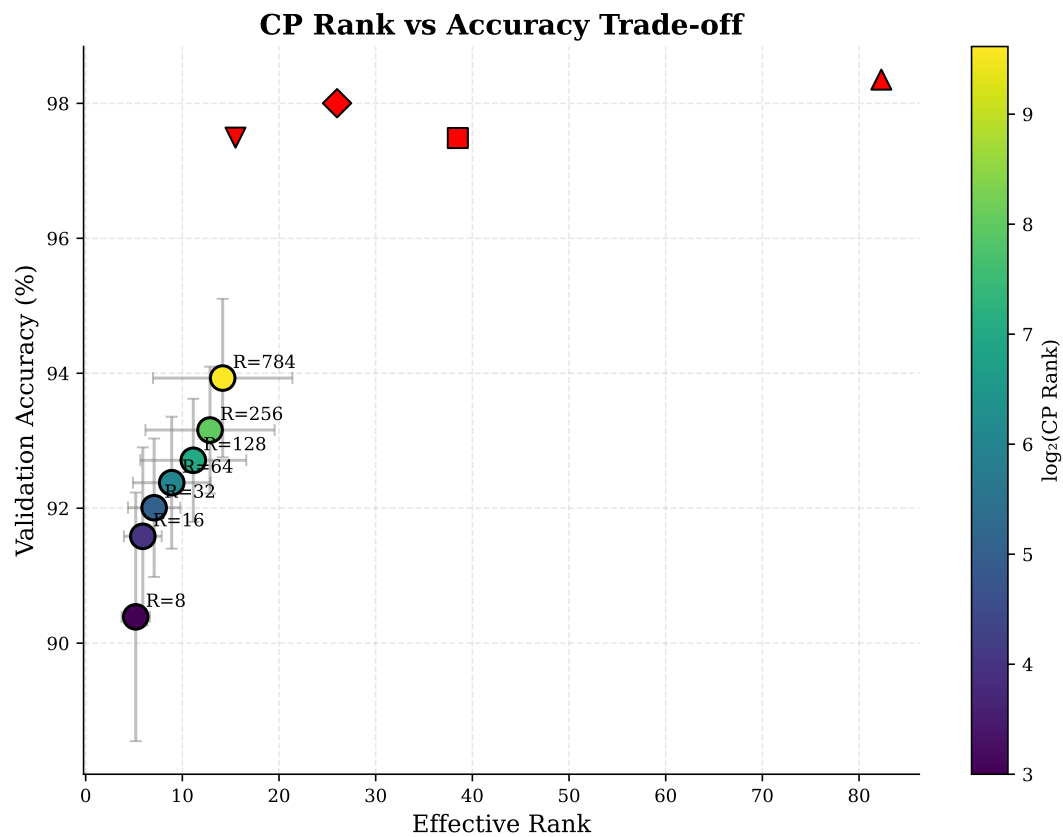


Figure 9: **Accuracy-Interpretability Pareto Frontier.** Fixed CP forces extreme sparsity (eff. rank ~ 3.5) at accuracy cost. Lambda and Gated CP maintain competitive accuracy while reducing effective rank.



Figure 10: **Feature Disentanglement: Dense vs. CP.** Top 5 eigenvectors for digits 0–9. **Dense:** Holistic, superimposed templates mixing multiple strokes. **CP:** Atomic, localized factors (isolated curves, specific strokes) that can be combined to reconstruct digits.

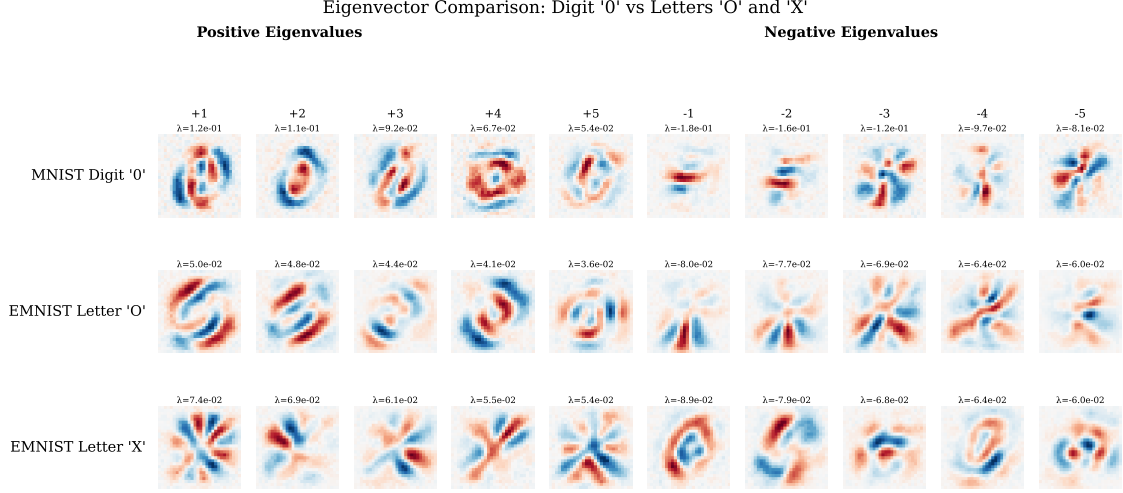


Figure 11: Three-way eigenvector comparison: MNIST-0, EMNIST-O, and EMNIST-X. Despite visual similarity between 0 and O, Quadratic Form Similarity successfully distinguishes them from the dissimilar X.

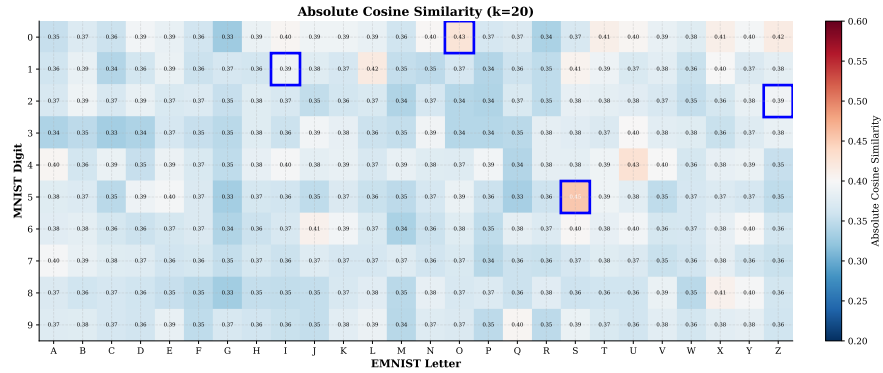


Figure 12: Absolute cosine similarity between MNIST digit classes and EMNIST letter classes fails to distinguish geometrically similar pairs (0-O) from dissimilar pairs (0-X).

figures/language/figure_8_side_by_side_comparison.pdf

Figure 13: Side-by-side comparison of AND-gate circuits. **Top row:** Feature 3834 (tutorial example) shows weak AND-gate structure with diffuse interactions. **Bottom row:** Feature 751 (best discovered) shows strong AND-gate structure with clear cluster separation and dominant opposing eigenvalues. Each row shows: interaction submatrix (left), eigenvector projections (middle), eigenvalue spectrum (right).